Pakka Agent — Production Environment & Crash Audit

Role: Senior Backend Engineer & Reliability Lead (Dev 2) | **Audience:** Raja & Team | **Target:** v1.0.0-rc1

A. Executive Summary

A rigorous, read-only audit of the entire Pakka Agent repository was conducted, spanning backend initialization (backend/src/server/main.ts), environment schema parsing (backend/src/lib/env.ts), Next.js frontend configuration, Railway deployment specifications (railway.json), Supabase RLS policies (migrations 0001-0052), and 28 Inngest background functions. The codebase typechecks cleanly and passes all 421 core tests, but **critical credential omissions in environment files will cause immediate boot crashes and payment failures if pushed directly to production without remediation.**

Audit Dimension	Status	Key Finding / Production Impact
Process Boot (Railway)	**CRITICAL RISK**	ANTHROPIC_API_KEY commented in local env; DB pre-sync dependency causes boot crash if DB unreachable.
Browser Client (Vercel)	**BROKEN (DEV)**	NEXT_PUBLIC_API_URL missing in frontend env; all browser API calls throw missing URL error.
Payments (PayU)	**TEST MODE ONLY**	Runs on PayU Sandbox (`PAYU_MODE=test`); live key/salt and auto-debit SI mandates pending from Raja.
WhatsApp Integration	**READY FOR LIVE**	Webhook handshake & HMAC-SHA256 signature verification cryptographically verified; needs Meta app link.
Inngest Background Jobs	**28/28 TESTED**	All 8 crons & 20 event workflows tested; production endpoint sync required on deploy.
Redis & Caching	**NOT USED**	Zero Redis dependencies in codebase; Postgres locks and Inngest durable queues handle state.
Token Encryption	**UNVERIFIED KEY**	AES-256-GCM encryption active; distinct 32-byte production key must be created via openssl.

B. Environment Variable Inventory & Classification

Every environment variable referenced across the frontend and backend runtime execution paths:

Variable	Required For	Host	Scope	Current Status
`NEXT_PUBLIC_APP_URL`	Frontend origin & CORS check	Railway & Vercel	Public	**PRESENT (UNVERIFIED)**
`NEXT_PUBLIC_API_URL`	Browser API caller base URL	Vercel	Public	**MISSING IN FRONTEND ENV**
`API_PUBLIC_URL`	Public API origin for PayU return	Railway	Public	**REQUIRED FOR BILLING**
`CORS_ALLOWED_ORIGINS`	Extra origins whitelist	Railway	Public	**PRESENT & VERIFIED**
`NEXT_PUBLIC_SUPABASE_URL`	Supabase project REST/Auth URL	Both	Public	**PRESENT & VERIFIED**
`NEXT_PUBLIC_SUPABASE_ANON_KEY`	Client Auth & RLS queries	Both	Public	**PRESENT & VERIFIED**
`SUPABASE_SERVICE_ROLE_KEY`	DB Admin, Storage, Background	Railway (ONLY)	SECRET	**LEAKED IN FRONTEND ENV**
`ANTHROPIC_API_KEY`	Claude 3.5 Sonnet & Haiku	Railway	SECRET	**COMMENTED OUT / CRASH**
`EMBEDDINGS_API_KEY`	Voyage-4 Vector Embeddings	Railway	SECRET	**PRESENT & VERIFIED**
`EMBEDDINGS_BASE_URL`	Atlas endpoint (voyage-4)	Railway	Public	**PRESENT & VERIFIED**
`ENCRYPTION_KEY`	AES-256-GCM tokens at rest	Railway (ONLY)	SECRET	**PRESENT (NEEDS PROD KEY)**
`PAYU_MERCHANT_KEY`	PayU Merchant identifier	Railway	Public	**TEST KEY PRESENT**
`PAYU_MERCHANT_SALT`	PayU SHA-512 signing salt	Railway	SECRET	**TEST SALT PRESENT**
`PAYU_MODE`	Switch ('test' vs 'live')	Railway	Public	**PRESENT ('test')**
`META_APP_SECRET`	Webhook HMAC-SHA256 sign	Railway	SECRET	**PRESENT & VERIFIED**
`META_WEBHOOK_VERIFY_TOKEN`	GET challenge verification	Railway	SECRET	**PRESENT & VERIFIED**
`META_SYSTEM_USER_TOKEN`	Meta Graph API Admin token	Railway	SECRET	**PRESENT (UNVERIFIED)**

WHATSAPP_DEMO_WABA_ID	Demo WABA Account ID	Railway	Public	PRESENT
WHATSAPP_DEMO_PHONE_NUMBER_ID	Demo WhatsApp Phone ID	Railway	Public	PRESENT
GOOGLE_CLIENT_ID / SECRET	Staff Google Calendar OAuth	Railway	SECRET	MISSING (EMPTY STRINGS)
GOOGLE_REDIRECT_URI	OAuth callback endpoint	Railway	Public	MISCONFIGURED (LOCALHOST)
INNGEST_EVENT_KEY	Inngest event dispatcher key	Railway	SECRET	PRESENT
INNGEST_SIGNING_KEY	Inngest webhook signature key	Railway	SECRET	PRESENT
INNGEST_SERVE_ORIGIN	Inngest callback host override	Railway	Public	MISSING (RECOMMENDED)
SENTRY_DSN	Direct envelope error reporting	Railway	Public	OPTIONAL / UNSET
LANGFUSE_PUBLIC_KEY / SECRET	LLM tracing & token counts	Railway	SECRET	OPTIONAL / UNSET
RESEND_API_KEY	Transactional email dispatch	Railway	SECRET	NOT REQUIRED (DAY 9)
REDIS_URL	In-memory cache/queues	None	N/A	NOT USED IN REPO

C. Crash and Error Report

Issue & Symptom	Root Cause	Sev	Affected Area	Required Fix
Backend Boot Crash EnvError: ANTHROPIC_API_KEY is required	Variable commented out in backend/.env.local line 16.	P0	Backend API startup	Uncomment & provide key in local; ensure set in Railway.
Client API Rejection NEXT_PUBLIC_API_URL is required	Omitted from frontend/.env.local; parseApiUrl throws on blank.	P1	All dashboard data reads	Set NEXT_PUBLIC_API_URL on Vercel to Railway domain.
Payment Checkout Refusal 503 not_available: Payments aren't switched on	checkout.ts:67 demands API_PUBLIC_URL to build return path.	P1	Plan upgrade & top-ups	Add API_PUBLIC_URL=https://api.pakkaagent.in to Railway.
Deploy Timeout Risk Railway healthcheck 60s timeout	syncPacks() blocks server listen before DB is confirmed ready.	P1	Deployment availability	Ensure DB is online before Railway deploy; migrations run first.
Google Calendar Connect Error 503 not_available: Google Calendar isn't on	GOOGLE_CLIENT_ID is empty string in config.	P2	Staff Calendar Sync	Configure verified Google OAuth client ID & secret.
Secret Exposure in Frontend Dir Service role key in frontend env	Server secrets pasted into frontend/.env.local.	P1	Security & credential leakage	Purge all non-NEXT_PUBLIC_* variables from frontend folder.

D. Third-Party Integration Deep-Dive

1. PayU India (Decision 20, Migration 0035)

• **Mechanism:** Auto-debit mandates (recurring SI) for subscriptions; one-time payments for top-ups (■1,199).
• **Security:** Inbound payment returns to /api/billing/payu/return are never trusted directly; the server executes a server-to-server verifyPayment() check signed with PAYU_MERCHANT_SALT.
• **Status:** Ready in test mode. Switch to PAYU_MODE=live requires Raja's merchant KYC and SI activation.
• **Launch Blocker:** YES (for paid conversions).

2. Meta WhatsApp Business Cloud API

• **Mechanism:** Inbound webhooks received at /api/webhooks/whatsapp; outbound messages via Meta Graph v26.0.
• **Security:** GET handshake verifies META_WEBHOOK_VERIFY_TOKEN with timing-safe SHA-256 compare. POST messages verify HMAC-SHA256 signature against META_APP_SECRET.
• **Status:** Configured with shared demo number (WABA 2454480031626919). Embedded Signup remains toggled off until Meta Tech Provider approval.
• **Launch Blocker:** YES.

3. Inngest (Durable Jobs Engine)

• **Mechanism:** Single endpoint at /api/inngest serving 28 functions (8 crons, 20 event workflows).
• **Security:** In production, rejects un-signed requests with HTTP 401 using INNGEST_SIGNING_KEY.
• **Status:** Working; requires explicit sync command (curl -X PUT https://api.pakkaagent.in/api/inngest) after deployment.
• **Launch Blocker:** YES.

4. Supabase (PostgreSQL 17, Auth, Storage, Realtime)

• **Database:** 51 migrations (0001 to 0052) cleanly applied. Strict RLS enforced across all tenant tables.
• **Storage:** Media stored in private bucket message-media accessed via service-role client. No separate S3 credentials required.
• **Status:** Staging operational on Mumbai (ap-south-1). Pro plan upgrade required to ensure automated daily backups and zero-pause.
• **Launch Blocker:** YES.

5. Sentry & Langfuse (Observability)

• **Sentry:** Direct envelope dispatcher (report.ts, no heavy SDK) with automated PII masking. Fails closed when unset.
• **Langfuse:** Optional LLM tracing. LANGFUSE_CAPTURE_TEXT defaults to false to prevent customer text leakage.
• **Launch Blocker:** NO (Observability enhancements).

E. Action Items for Dev 2

Category	Action Item	Target Location	Required Access
Repository	Add NEXT_PUBLIC_API_URL to frontend env; purge service-role secrets from frontend folder.	frontend/.env.local	Local Repo
Railway	Set API_PUBLIC_URL, ANTHROPIC_API_KEY, and generate fresh 32-byte ENCRYPTION_KEY.	Railway Dashboard	Railway Production
Vercel	Set NEXT_PUBLIC_API_URL, Supabase URLs, and Meta App ID. Confirm no secrets exist.	Vercel Dashboard	Vercel Project
Supabase	Push migrations 0001-0052; configure Auth Site URL to https://app.pakkaagent.in.	Supabase Console	Supabase Admin
Inngest	Execute curl -X PUT https://api.pakkaagent.in/api/inngest and confirm 28 functions registered.	Inngest Cloud	Inngest Org
Raja / PayU	Obtain live PayU Merchant Key & Salt; register webhook URL /api/webhooks/payu.	PayU Merchant Portal	Raja (Owner)

F. Prioritized Execution Plan

Phase 1: P0 (Boot Blockers — Fix Immediately)
1. Populate ANTHROPIC_API_KEY, EMBEDDINGS_API_KEY, and ENCRYPTION_KEY in Railway.
2. Add NEXT_PUBLIC_API_URL to Vercel and API_PUBLIC_URL to Railway.
3. Verify Railway healthcheck answers HTTP 200 on /api/health.

Phase 2: P1 (Pre-Launch Functional Verification)
4. Apply full migration suite (0001 to 0052) on production Supabase.
5. Re-sync Inngest production app and verify 28 functions show green.
6. Test PayU test transaction drill (■1,199 top-up) and verify credit grant.
7. Complete Meta webhook challenge handshake on live domain.

Phase 3: P2 (Operational Hardening)
8. Configure Sentry DSN and verify alert thresholds.
9. Execute weekly database backup dump drill to encrypted storage.
10. Connect verified Google Cloud OAuth credentials for staff Google Calendar sync.